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Oil Market Simulation Model

User's Manual

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MASTER

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Oil Market Simulation Model User's Manual

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Preface

The Oil Market Simulation (OMS) model is one of more than 30 analytic computer models operated by the Energy Information Administration (EIA). Because they were designed to run in a mainframe environment, these models are available to the public only in 9-track form (through the National Technical Information Service).

The increasingly heavy use of personal computers by analysts working in all sectors of the economy has prompted EIA to consider a program to compress its models to spreadsheet size. OMS was chosen as a test case because of widespread interest in this model and its suitability for conversion to a spreadsheet format.

The OMS spreadsheet version is a cooperative project between W. Calvin Kilgore, Director of Energy Markets and End Use and Douglas Hale of the Office of Statistical Standards. Questions may be directed to Derriel Cato (586-6574) or Erik A. Kreil (586-6573), in the Office of Energy Markets and End Use.

This package of two diskettes and one user's manual is available from either of the following sources:

National Technical Information Service (NTIS)
Office of Data Base Services
U.S. Department of Commerce
5285 Port Royal Road
Springfield, VA 22161
(703)487-4650

National Energy Information Center (NEIC), EI-231
Energy Information Administration
Forrestal Building
Room 1F-048
Washington, DC 20585
(202) 586-8800

If you are interested in studying the mathematics of the system in detail, you will have to read the documentation of the parent (mainframe) version. The two EIA documentation reports referenced on page 2 of this manual are available on microfiche for a marginal fee from NTIS or in paper form free from NEIC.

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Oil Market Simulation Model User's Manual

The Oil Market Simulation (OMS) Model is a LOTUS 1-2-3 Spreadsheet that simulates the world oil market. OMS is an annual model with a data base that begins in 1979 and computes projections through the year 2000. The geographic coverage includes all market economies, with net imports from the centrally planned economies taken as an assumption. The model estimates the effects of price changes on oil supply and demand and computes an oil price path over time that allows supply and demand to remain in balance within the market economies area as a whole.

The input assumptions of OMS are highlighted (in color) on the spreadsheet and include the following:

- The capacity of the OPEC countries to produce petroleum liquids (crude oil, natural gas liquids, condensates, refinery gains);
- A reference case projection of regional oil supply and demand at some arbitrary reference path of oil prices over time. The reference case provided with this diskette is that used for EIA's latest base case in the *International Energy Outlook, 1987* DOE/EIA-0484(87).

Using the Demonstration Diskette

The demonstration requires an IBM PC (or compatible), preferably with a color monitor. The demonstration diskette is self-contained, with all the files needed to run the demonstration. It does not, however, have the DOS system files, so this diskette cannot be used to start (boot) the computer. To view the demonstration, the user should do the following:

Start the computer as is normally done. From the DOS prompt, usually **A>**, the user should put the demonstration diskette in disk drive A and type

DEMO <enter>

This is a simple batch file with two parts: one to run the demonstration of the text screens, and one to run a demonstration of the graphs. (In between the two parts, the computer may require the DOS diskette to be inserted temporarily into the disk drive to re-load the **COMMAND.COM** file.)

Using the OMS Model

This model requires Release 2 of LOTUS' 1-2-3. Start LOTUS' 1-2-3 as is usually done. Load the worksheet (using / *File Retrieve*), which is called **OMS.WK1**. It is large and will take a while to load. Once loaded, it will automatically set up the Alternative Menu, which allows easy access to all the features of the OMS Model. This menu works in the same way as the standard 1-2-3 menu: select an option either by the first letter of the option or by "point-and-shoot" (cursor keys to the option and then hit **ENTER**).

Simulation Procedure

There are three ways in which a simulation can be run:

1. Evaluate a given price path for all years of the model. The demand for OPEC oil is the difference between total demand and non-OPEC supply.
2. Use OPEC's price-reaction function for some or all years of the model. In a given year, this implies a price at which the demand for OPEC oil intersects OPEC's price-reaction function. If this function is used for only some interval of years, the "active" price path is used for years outside this interval, with the demand for OPEC oil in those years being the difference between total demand and non-OPEC supply at the given price.
3. Evaluate a given output path for OPEC, for some or all years of the model. A market-clearing price for each year, as well as demand and non-OPEC supply quantities, are calculated such that the difference between total demand and non-OPEC supply is exactly equal to the given OPEC output for that year. If an output path is specified for only some interval of years, the "active" price path is used for years outside this interval, with the demand for OPEC oil in those years being the difference between total demand and non-OPEC supply at the given price.

OMS Model Documentation

OMS has been extensively used in EIA analyses since 1978. In addition to generating oil market projections for every issue of the Agency's annual reports to Congress, the model has been used in numerous interagency studies over the years, including the recently completed *Energy Security A Report to the President of the United States*, March 1987, DOE/S-0057.

OMS is intended to mimic the behavior of several national and regional models which collectively forecast the world oil price, regional supplies, and regional demands. Presented with an initial projection from these models, OMS estimates how the initial projection would change in response to new conditions. The initial projections are called reference values. Though the purpose of OMS is to project changes, it implicitly

projects levels as the product of the reference values and the change projections by OMS. Since OMS calculates estimates much more quickly and cheaply than the original models, it offers potential savings over the larger models in sensitivity analyses.

The OMS model has had extensive documentation for its mainframe version. The previous documentation provides additional information on the concepts and variables underlying the LOTUS 1-2-3 version. There is no difference in the methodology used in the PC and mainframe versions. The main difference lies in whether LOTUS 1-2-3 macros or FORTRAN subroutines are used to perform the calculations.

- *Oil Market Simulation Model Documentation Report* June 1983 DOE/EIA-0412
- *The Oil Market Simulation Model Model Documentation Report* May 1985 DOE/EI/19656-2.

Figure 1. OMS Model

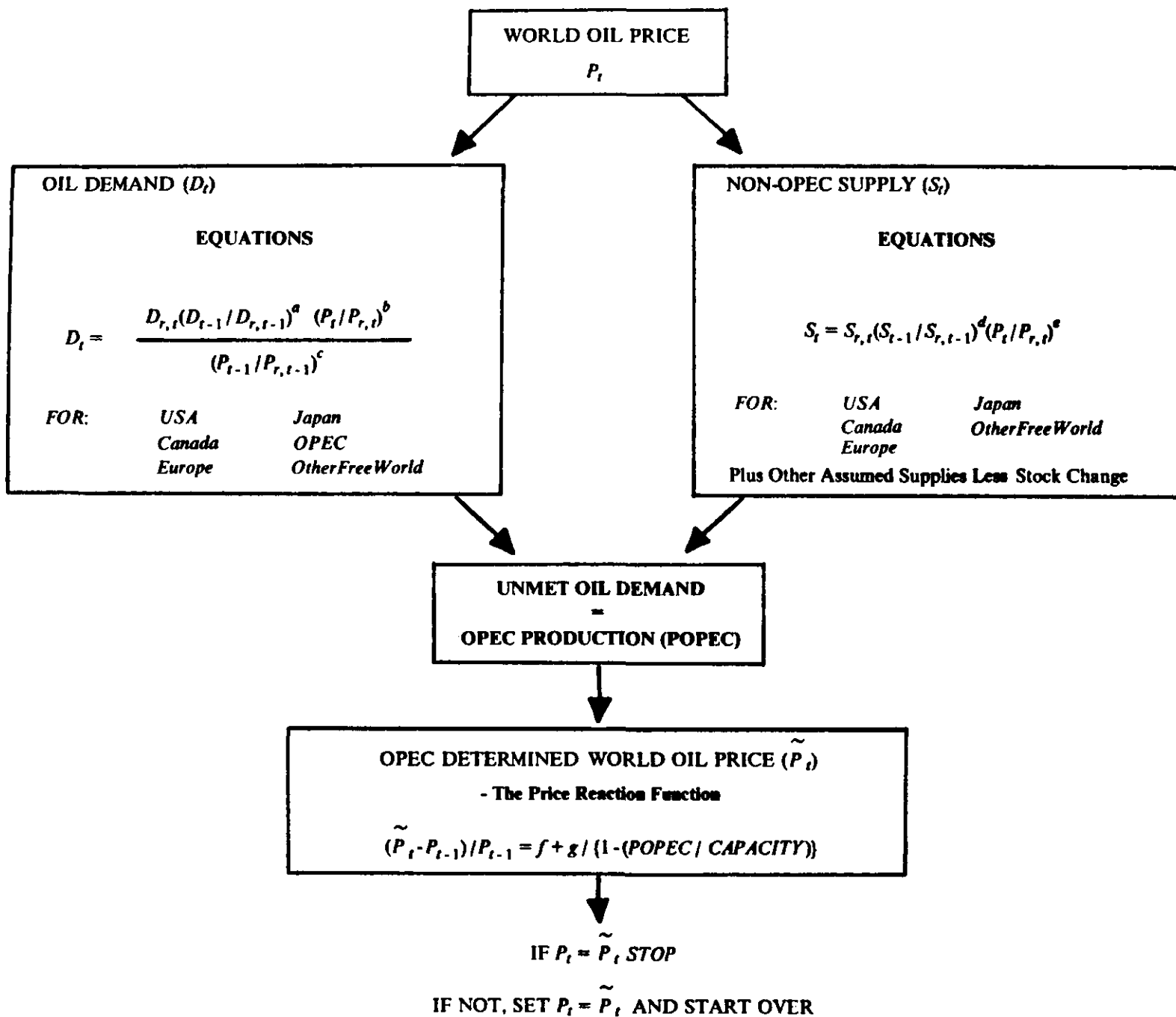


Table 1. Oil Market Simulation Model Data Assumptions

Data Type	Location in Spreadsheet					
	United States	Canada	Japan	OECD Europe	Other Non-OPEC	OPEC
Historical Demand (D_t)	C17..I17	C33..I33	C49..I49	C65..I65	C81..I81	C97..I97
Reference Demand ($D_{r,t}$)	J18..X18	J34..X34	J50..X50	J66..X66	J82..X82	J97..X97
Historical Supply (S_t)	C20..I20	C36..I36	C52..I52	C68..I68	C84..I84	C105..I105
Reference Supply ($S_{r,t}$)	J21..X21	J37..X37	J53..X53	J69..X69	J85..X85	-
Supply Lag (d)	B26	B42	B58	B74	B90	-
Demand Lag (a)	B27	B43	B59	B75	B91	-
Income Elasticity (γ)	B28	B44	B60	B76	B92	-
Demand Elasticity (β)	H29..X29	H45..X45	H61..X61	H77..X77	H93..X93	-
Feedback Elasticity (α)	H30..X30	H46..X46	H62..X62	H78..X78	H94..X94	-
Supply Elasticity (e)	H31..X31	H47..X47	H63..X63	H79..X79	H95..X95	-
Production Capacity (Capacity)	-	-	-	-	-	C8..X8
Historical Prices (P)	C11..I11					
Reference Prices (P_{rt})	J12..X12					
Price Behavior (f and g)	B14..B15					

The model's design abstracts from the complexity of the underlying country and regional models by limiting its scope and by making strong assumptions about markets. To limit its scope, OMS:

- Considers OPEC and only five other supply and demand regions, and,
- Forecasts only 12 quantities over time.

To simplify the analysis of markets, OMS assumes that:

- The current oil price and last year's supplies and demands are the only determinants of non-OPEC, non-Communist supply and demand
- Other factors are implicit in the reference supplies and demands
- Price taking behavior by all countries and regions except for OPEC
- OPEC follows a mechanical pricing rule, called the price-reaction function, to increase price with increased capacity utilization.

These assumptions result in a model consisting of only 12 equations (5 supply, 5 demand, OPEC production, and world price) and 2 identities. Figure 1 shows the general structure of the OMS Model. In the figure, the subscript r refers to the assumed reference value and t and $t-1$ refer to this year and last year, respectively. Parameters are indicated by a , b , c , d , e , f , and g ; the latter two are estimated from market data. Capacity is OPEC's maximum sustainable production capacity.

OMS projects a market clearing price for each year. At the market clearing price, world oil demand equals the sum of oil production from all sources, including stock changes. Stock changes are assumed. Simultaneously, OPEC is meeting its price objective for the oil it produces. To find the market clearing price, the model starts with an arbitrary trial world oil price, shown as $P(t)$ at the top of Figure 1.

Non-OPEC demands and supplies at that price are then calculated according to the formulae shown in the second row of boxes. The excess of demand over supply is assumed to be met by OPEC production. The price-reaction curve, shown in the lowest box in Figure 1, yields the price OPEC would charge to produce that quantity of oil. If the calculated price is not the same as the trial price, the process starts over again with the calculated price. If the calculated price equals the trial price, the market clears at that price.

Reference Data, Parameters, and Equations

The equations used in OMS are shown in Figure 1. In practice, the demand equation coefficient b is decom-

posed into a price elasticity of demand (β) plus the product of a feedback (α) and an income (γ) elasticity. That is, $b = \beta + \alpha\gamma$. Similarly, the demand equation coefficient c is the product of feedback and income elasticities and a lag coefficient. That is $c = \alpha\gamma\alpha$.

The location of the data and parameters on the LOTUS Spreadsheet are given in Table 1.

A graph of the OPEC price reaction function is provided in Figure 2.

Substituting the values provided in the spreadsheet into the equations displayed in Figure 2 yields all of the non-OPEC supply and Free World demand curves, and the price reaction function. The system is closed by the requirement that unmet oil demand equals OPEC production at equilibrium

(i.e., $P_t = P_t$).

Unmet oil demand is the sum of regional oil demands less the sum of non-OPEC production, net imports from Communist countries, and stock withdrawals.

Figure 2. OPEC Price Reaction Function

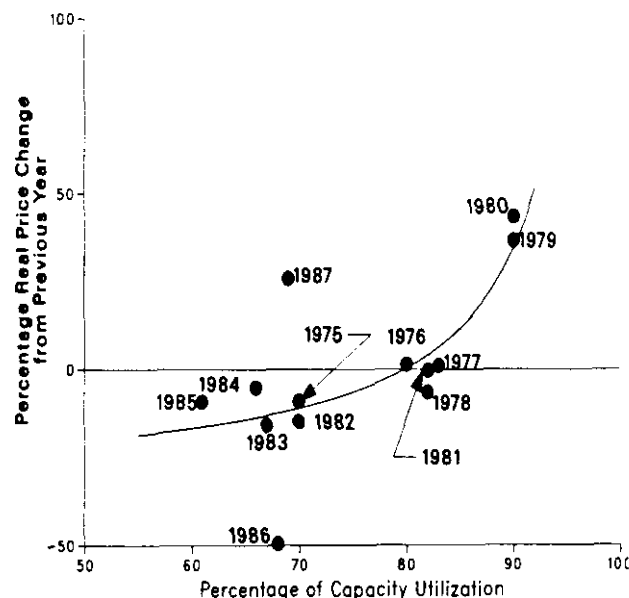


Table 2. Data for Estimating Price Reaction Function

Year	OPEC ¹ Capacity (MMBD)	OPEC ² Production (MMBD)	WorldOil ³ Price (\$ 1986)
1974	38.6	31.182	26.45
1975	39.3	27.645	24.05
1976	39.0	31.297	24.38
1977	38.3	31.901	24.63
1978	37.3	30.537	23.03
1979	35.2	31.759	31.46
1980	30.8	27.809	45.12
1981	29.0	23.726	44.97
1982	28.5	19.938	38.28
1983	28.0	18.635	32.21
1984	28.2	18.652	30.54
1985	28.4	17.385	27.69
1986	28.9	19.743	13.98
1987	28.4	19.471	17.59

¹Production capacity is an EIA estimate. It includes natural gas liquids production, processing gain, and other liquids as well as crude oil.

²Production is from EIA's "International Petroleum Statistics Report," March 25, 1988. It includes natural gas liquids production, processing gain, and other liquids as well as crude oil.

³The U.S. imported refiner acquisition cost. The nominal dollar series is from the *1986 Annual Energy Review*, except for 1987, where \$18.11 is used. The 1975 nominal price excludes a \$1.43 import fee.

Parameter Estimation

The parameters of the price reaction function were estimated by Ordinary Least Squares. The data used for estimating the price reaction function is shown in Table 2.

The non-OPEC oil demand and supply parameters were estimated by "calibrating" OMS to the most recent energy market projections published by the Energy Information Administration. These projections are updated annually and documented in the *Annual Energy Outlook* and the *International Energy Outlook*.

The domestic projections are projected with the aid of the Intermediate Future Forecasting System (IFFS) and its many submodels on specific energy sectors. The international projections are based on similar anal-

yses of energy supply and demand in the other regions of the world and accumulated in LOTUS 1-2-3 spreadsheets to construct overall world energy balances.

Based upon EIA's annual update of its world energy projections, a complete set of projected future world oil demand and non-OPEC supply levels at some assumed oil price level is input to OMS. The supply and demand elasticities and lag parameters in OMS are then specified to exactly duplicate EIA's latest base case projections. The parameters are also estimated to approximate, as closely as possible, the sensitivities of these base case projections to changes in oil prices as depicted in the various sensitivity cases that are also shown in EIA's annual energy projections.

Solving the Model

The solution algorithm in the model solves for the price at which the demand for OPEC oil (total demand less non-OPEC supply) intersects the price-reaction function. That is, the model searches to find a price P^* at which total demand $D = f(P)$ less non-OPEC supply $S = g(P)$ equals the level of OPEC output from (the function which is the inverse to) the price-reaction function $X = h^{-1}(P)$.

In the case of a given OPEC output path, the solution is the price at which the demand for OPEC oil intersects the fixed level of OPEC output X' : the price P' at which total demand $D = f(P)$, less non-OPEC supply, $S = g(P)$, equals the given level of OPEC output X' .

The algorithm uses a standard iterating procedure, the Newton-Raphson algorithm, to find the solution price, which is just the root of the function:

$$f(P) - g(P) = h^{-1}(P) \quad (1)$$

or, for given OPEC output, the function:

$$f(P) - g(P) = X'. \quad (2)$$

Each iteration gets closer and closer to the solution, by adjusting the current estimate of the solution price up or down according to the respective function's first derivative. It stops searching when the next adjustment to the price would be less than a half cent. Such solution techniques are discussed in *Mathematical Applications of Electronic Spreadsheets* by Deane E. Arganbright (McGraw Hill, 1985).